# PROJECT

## -- PRODUCTS

```
-- Basic Exploration PRODUCTS
------------------------------------------------------------
-- Preview first 10 rows
SELECT *
FROM SH.PRODUCTS
FETCH FIRST 10 ROWS ONLY;
```

OUTPUT

| PROD_ID | PROD_NAME | PROD_DESC | PROD_SUBCATEGOI | PROD_SUBCATEGOI | PROD_SUBCATEGOI | PROD_CATEGORY | PROD_CATEGORY_I | PROD_CATEGORY_I | PROD_WEIGHT_CLA | PI |
|---|---|---|---|---|---|---|---|---|---|---|
| 14 | Pitching Machine anc | Pitching Machine anc | Training Aids and Eq | 2035 | Training Aids and Eq | Baseball | 203 | Baseball | 1 | U |
| 21 | Speed Trainer Bats a | Speed Trainer Bats a | Training Aids and Eq | 2035 | Training Aids and Eq | Baseball | 203 | Baseball | 1 | U |
| 22 | MLB Official Game B | MLB Official Game B | Baseballs | 2031 | Baseballs | Baseball | 203 | Baseball | 1 | U |
| 46 | 2 Competition Grade | 2 Competition Grade | Baseballs | 2031 | Baseballs | Baseball | 203 | Baseball | 1 | U |
| 47 | 6 Gallon Empty Ball E | 6 Gallon Empty Ball E | Baseballs | 2031 | Baseballs | Baseball | 203 | Baseball | 1 | U |
| 127 | Genuine Series MIX V | Genuine Series MIX V | Bats | 2036 | Bats | Baseball | 203 | Baseball | 1 | U |
| 128 | Slugger Youth Series | Slugger Youth Series | Bats | 2036 | Bats | Baseball | 203 | Baseball | 1 | U |
| 129 | Pro Maple Bat | Pro Maple Bat | Bats | 2036 | Bats | Baseball | 203 | Baseball | 1 | U |
| 130 | Pro Maple Youth Bat | Pro Maple Youth Bat | Bats | 2036 | Bats | Baseball | 203 | Baseball | 1 | U |
| 19 | Cricket Bat Bag | Cricket bat bag | Cricket Bat | 2051 | Cricket Bat | Cricket | 205 | Cricket | 1 | U |

```
-- Count total products
SELECT COUNT(*) AS total_products
FROM SH.PRODUCTS;
```

OUTPUT

| | TOTAL_PRODUCTS |
|---|---|
| 1 | 72 |

```
-- Count distinct product IDs
SELECT COUNT(DISTINCT PROD_ID) AS distinct_product_ids
FROM SH.PRODUCTS;
```

OUTPUT

| | DISTINCT_PRODUCT_IDS |
|---|---|
| 1 | 72 |

```
-- Products grouped by category
SELECT PROD_CATEGORY, COUNT(*) AS product_count
FROM SH.PRODUCTS
GROUP BY PROD_CATEGORY
ORDER BY product_count DESC;
```

OUTPUT

| | PROD_CATEGORY | PRODUCT_COUNT |
|---|---|---|
| 1 | Cricket | 26 |
| 2 | Baseball | 21 |
| 3 | Tennis | 13 |
| 4 | Soccer / Football | 10 |
| 5 | Golf | 2 |

```sql
-- Products grouped by subcategory
SELECT PROD_SUBCATEGORY, COUNT(*) AS product_count
FROM SH.PRODUCTS
GROUP BY PROD_SUBCATEGORY
ORDER BY product_count DESC;
```

OUTPUT

| | PROD_SUBCATEGOI | PRODUCT_COUNT |
|---|---|---|
| 1 | Cricket | 8 |
| 2 | Cricket Clothing | 7 |
| 3 | Cricket Fan Gear | 6 |
| 4 | Gloves and Mitts | 6 |
| 5 | Baseballs | 5 |
| 6 | Soccer Equipment | 5 |
| 7 | Cricket Bat | 5 |
| 8 | Bats | 4 |
| 9 | Tennis Strings | 4 |
| 10 | Tennis Balls | 4 |
| 11 | Soccer Clothing | 3 |

| | PROD_SUBCATEGOI | PRODUCT_COUNT |
|---|---|---|
| 9 | Tennis Strings | 4 |
| 10 | Tennis Balls | 4 |
| 11 | Soccer Clothing | 3 |
| 12 | Tennis Racquet | 3 |
| 13 | Batting Tees | 2 |
| 14 | Baseball Clothing | 2 |
| 15 | Soccer / Football | 2 |
| 16 | Training Aids and Eq | 2 |
| 17 | Tennis Racquet Grip | 2 |
| 18 | Golf Bags and Carts | 1 |
| 19 | Irons and Iron Sets | 1 |

```
-- Check for missing values in key columns
SELECT
    SUM(CASE WHEN PROD_NAME IS NULL THEN 1 ELSE 0 END) AS null_prod_name,
    SUM(CASE WHEN PROD_CATEGORY IS NULL THEN 1 ELSE 0 END) AS null_prod_category,
    SUM(CASE WHEN PROD_LIST_PRICE IS NULL THEN 1 ELSE 0 END) AS null_list_price,
    SUM(CASE WHEN PROD_STATUS IS NULL THEN 1 ELSE 0 END) AS null_prod_status
FROM SH.PRODUCTS;
```

OUTPUT

| | NULL_PROD_NAME | NULL_PROD_CATEGORY | NULL_LIST_PRICE | NULL_PROD_STATUS |
|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 |

```
-- Insights
------------------------------------------------------------
-- Average price per category
SELECT PROD_CATEGORY, AVG(PROD_LIST_PRICE) AS avg_price
FROM SH.PRODUCTS
GROUP BY PROD_CATEGORY
ORDER BY avg_price DESC;
```

OUTPUT

| | PROD_CATEGORY | AVG_PRICE |
|---|---|---|
| 1 | Golf | 1149.99 |
| 2 | Soccer / Football | 230.19 |
| 3 | Baseball | 145.08523809523808 |
| 4 | Tennis | 121.68230769230769 |
| 5 | Cricket | 31.413076923076922 |

```
-- Top 10 most expensive products
SELECT PROD_ID, PROD_NAME, PROD_LIST_PRICE
FROM SH.PRODUCTS
ORDER BY PROD_LIST_PRICE DESC
FETCH FIRST 10 ROWS ONLY;
```

OUTPUT

| | PROD_ID | PROD_NAME | PROD_LIST_PRICE |
|---|---|---|---|
| 1 | 18 | Lithium Electric Golf Caddy | 1299.99 |
| 2 | 17 | Soccer Goal - Official | 1099.99 |
| 3 | 14 | Pitching Machine and Batting Cage Combo | 999.99 |
| 4 | 15 | Right-Handed Graphite Shaft Iron Set | 999.99 |
| 5 | 21 | Speed Trainer Bats and Training Program | 899.99 |
| 6 | 13 | Soccer Goal - Portable | 899.99 |
| 7 | 20 | Match Used Autograph Racquet | 599.99 |
| 8 | 29 | Limited Edition Racquet | 499.99 |
| 9 | 16 | Sonic Core Graphite Racquet | 299.99 |
| 10 | 28 | English Willow Cricket Bat | 199.99 |

```
-- Products with largest gap between list price and minimum price
SELECT PROD_ID, PROD_NAME, PROD_LIST_PRICE, PROD_MIN_PRICE,
       (PROD_LIST_PRICE - PROD_MIN_PRICE) AS price_gap
FROM SH.PRODUCTS
ORDER BY price_gap DESC
FETCH FIRST 10 ROWS ONLY;
```

OUTPUT

| | PROD_ID | PROD_NAME | PROD_LIST_PRICE | PROD_MIN_PRICE | PRICE_GAP |
|---|---|---|---|---|---|
| 1 | 14 | Pitching Machine and Batting Cage Combo | 999.99 | 999.99 | 0 |
| 2 | 21 | Speed Trainer Bats and Training Program | 899.99 | 899.99 | 0 |
| 3 | 22 | MLB Official Game Baseball w/ Display Case | 24.99 | 24.99 | 0 |
| 4 | 46 | 2 Competition Grade NFHS Baseballs | 22.99 | 22.99 | 0 |
| 5 | 47 | 6 Gallon Empty Ball Bucket | 28.99 | 28.99 | 0 |
| 6 | 127 | Genuine Series MIX Wood Bat | 36.99 | 36.99 | 0 |
| 7 | 128 | Slugger Youth Series Maple Bat | 27.99 | 27.99 | 0 |
| 8 | 129 | Pro Maple Bat | 192.99 | 192.99 | 0 |
| 9 | 130 | Pro Maple Youth Bat | 89.99 | 89.99 | 0 |
| 10 | 19 | Cricket Bat Bag | 55.99 | 55.99 | 0 |

```
-- Relationships with Tables
------------------------------------------------------------
-- Sales by product category
SELECT p.PROD_CATEGORY, SUM(s.AMOUNT_SOLD) AS total_sales
FROM SH.SALES s
JOIN SH.PRODUCTS p ON s.PROD_ID = p.PROD_ID
GROUP BY p.PROD_CATEGORY
ORDER BY total_sales DESC;
```

OUTPUT

| | PROD_CATEGORY | TOTAL_SALES |
|---|---|---|
| 1 | Baseball | 31163988.41 |
| 2 | Golf | 20647605.6 |
| 3 | Soccer / Football | 17961865.9 |
| 4 | Tennis | 14597515.62 |
| 5 | Cricket | 13834855.68 |

```
-- Average unit cost vs unit price by product
SELECT p.PROD_NAME, AVG(c.UNIT_COST) AS avg_cost, AVG(c.UNIT_PRICE) AS avg_price
FROM SH.COSTS c
JOIN SH.PRODUCTS p ON c.PROD_ID = p.PROD_ID
GROUP BY p.PROD_NAME
ORDER BY avg_price DESC;
```

OUTPUT

| | PROD_NAME | AVG_COST | AVG_PRICE |
|---|---|---|---|
| 1 | Lithium Electric Golf Caddy | 1166.2701375814627 | 1523.840999275887 |
| 2 | Soccer Goal - Official | 995.0602578796562 | 1306.8822206303726 |
| 3 | Pitching Machine and Batting Cage Combo | 891.7257102272728 | 1175.4339962121212 |
| 4 | Speed Trainer Bats and Training Program | 843.7410763888889 | 1055.1727777777778 |
| 5 | Soccer Goal - Portable | 762.1709390862944 | 1026.2508629441625 |
| 6 | Right-Handed Graphite Shaft Iron Set | 844.9934897025172 | 969.8684782608696 |
| 7 | Match Used Autograph Racquet | 521.2212841530055 | 614.4891256830601 |
| 8 | Limited Edition Racquet | 445.8564791528789 | 538.0332097948378 |
| 9 | Sonic Core Graphite Racquet | 266.2917455621302 | 299.1811094674556 |
| 10 | English Willow Cricket Bat | 172.72197171381032 | 209.17896838602329 |

## -- CUSTOMERS

```
-- Basic Exploration CUSTOMERS
------------------------------------------------------------
-- Preview first 10 rows
SELECT *
FROM SH.CUSTOMERS
FETCH FIRST 10 ROWS ONLY;
```

OUTPUT

| | CUST_ID | CUST_FIRST_NAME | CUST_LAST_NAME | CUST_GENDER | CUST_YEAR_OF_BIF | CUST_MARITAL_ST/ | CUST_STREET_ADD | CUST_POSTAL_COD | CUST_CITY | CUST_CITY_ID | CI |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 16561 | Abigail | Grimes | M | 1954 | (null) | 67 Packard Avenue | 53574 | Almere | 51057 | Fl |
| 2 | 20116 | Abigail | Grimes | M | 1958 | single | 67 South Coshocton | 63488 | Sydney | 52334 | N |
| 3 | 23672 | Abigail | Grimes | M | 1976 | married | 77 West Sagadahoc / | 72620 | Enkhuizen | 51452 | N |
| 4 | 27227 | Abigail | Grimes | M | 1961 | (null) | 87 East Geneva Aven | 76836 | Amersfoort | 51062 | U |
| 5 | 30782 | Abigail | Grimes | M | 1982 | single | 87 South Toa Alta Av | 37400 | Yokohama | 52526 | K |
| 6 | 34339 | Abigail | Grimes | M | 1964 | married | 97 North Kanabec A | 69987 | Blumenau | 51201 | S |
| 7 | 37005 | Abigail | Husky | M | 1958 | married | 107 East Bradford Av | 33529 | Londrina | 51804 | P |
| 8 | 40560 | Abigail | Husky | M | 1973 | (null) | 107 South Packard A | 69987 | Blumenau | 51201 | S |
| 9 | 44115 | Abigail | Husky | M | 1992 | single | 117 North Coshocto | 81052 | Delft | 51383 | Z |
| 10 | 34348 | Abigail | Husky | M | 1965 | married | 17 North Kanabec C | 76486 | Buenos Aires | 51223 | B |

```
-- Count total customers
SELECT COUNT(*) AS total_customers
FROM SH.CUSTOMERS;
```

OUTPUT

| | TOTAL_CUSTOMERS |
|---|---|
| 1 | 55500 |

```
-- Count distinct customer IDs
SELECT COUNT(DISTINCT CUST_ID) AS distinct_customer_ids
```

```
FROM SH.CUSTOMERS;
```

OUTPUT

| | DISTINCT_CUSTOMER_IDS |
|---|---|
| 1 | 55500 |

```
-- Profiling
------------------------------------------------------------
-- Customers grouped by marital status
SELECT CUST_MARITAL_STATUS, COUNT(*) AS customer_count
FROM SH.CUSTOMERS
GROUP BY CUST_MARITAL_STATUS
ORDER BY customer_count DESC;
```

OUTPUT

| | CUST_MARITAL_STATUS | CUSTOMER_COUNT |
|---|---|---|
| 1 | single | 19094 |
| 2 | *(null)* | 17506 |
| 3 | married | 16287 |
| 4 | never married | 1503 |
| 5 | divorced | 765 |
| 6 | widowed | 211 |
| 7 | separated | 134 |

```
-- Customers grouped by income level
SELECT CUST_INCOME_LEVEL, COUNT(*) AS customer_count
FROM SH.CUSTOMERS
GROUP BY CUST_INCOME_LEVEL
ORDER BY customer_count DESC;
```

OUTPUT

| | CUST_INCOME_LEVEL | CUSTOMER_COUNT |
|---|---|---|
| 1 | F: 110,000 - 129,999 | 10537 |
| 2 | E: 90,000 - 109,999 | 7936 |
| 3 | G: 130,000 - 149,999 | 5465 |
| 4 | H: 150,000 - 169,999 | 5465 |
| 5 | D: 70,000 - 89,999 | 5217 |
| 6 | I: 170,000 - 189,999 | 4566 |
| 7 | C: 50,000 - 69,999 | 4255 |
| 8 | J: 190,000 - 249,999 | 3006 |
| 9 | B: 30,000 - 49,999 | 2764 |
| 10 | A: Below 30,000 | 2630 |
| 11 | K: 250,000 - 299,999 | 1934 |
| 12 | L: 300,000 and above | 1684 |
| 13 | *(null)* | 41 |

```
-- NULL Value Checks
SELECT
    SUM(CASE WHEN CUST_FIRST_NAME IS NULL THEN 1 ELSE 0 END) AS null_first_name,
    SUM(CASE WHEN CUST_LAST_NAME IS NULL THEN 1 ELSE 0 END) AS null_last_name,
    SUM(CASE WHEN CUST_EMAIL IS NULL THEN 1 ELSE 0 END) AS null_email,
    SUM(CASE WHEN CUST_INCOME_LEVEL IS NULL THEN 1 ELSE 0 END) AS null_income_level
FROM SH.CUSTOMERS;
```

OUTPUT

| | NULL_FIRST_NAME | NULL_LAST_NAME | NULL_EMAIL | NULL_INCOME_LEVEL |
|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 41 |

```
-- Replace missing income level with 'Unknown'
UPDATE SH.CUSTOMERS
SET CUST_INCOME_LEVEL = 'Unknown'
WHERE CUST_INCOME_LEVEL IS NULL;

-- Insights
------------------------------------------------------------
--Distribution by Gender
SELECT CUST_GENDER, COUNT(*) AS customer_count
FROM SH.CUSTOMERS
GROUP BY CUST_GENDER
ORDER BY customer_count DESC;
```

OUTPUT

| | CUST_GENDER | CUSTOMER_COUNT |
|---|---|---|
| 1 | M | 37175 |
| 2 | F | 18325 |

```sql
-- Age Segmentation
SELECT
    FLOOR((EXTRACT(YEAR FROM SYSDATE) - CUST_YEAR_OF_BIRTH)/10)*10 AS age_group,
    COUNT(*) AS customer_count
FROM SH.CUSTOMERS
GROUP BY FLOOR((EXTRACT(YEAR FROM SYSDATE) - CUST_YEAR_OF_BIRTH)/10)*10
ORDER BY age_group;
```

OUTPUT

| | AGE_GROUP | CUSTOMER_COUNT |
|---|---|---|
| 1 | 20 | 602 |
| 2 | 30 | 7845 |
| 3 | 40 | 9373 |
| 4 | 50 | 10631 |
| 5 | 60 | 14355 |
| 6 | 70 | 9090 |
| 7 | 80 | 2904 |
| 8 | 90 | 695 |
| 9 | 100 | 5 |

```sql
-- Average credit limit by income level
SELECT CUST_INCOME_LEVEL, AVG(CUST_CREDIT_LIMIT) AS avg_credit_limit
FROM SH.CUSTOMERS
GROUP BY CUST_INCOME_LEVEL
ORDER BY avg_credit_limit DESC;
```

OUTPUT

| | CUST_INCOME_LEVEL | AVG_CREDIT_LIMIT |
|---|---|---|
| 1 | K: 250,000 - 299,999 | 10540.847983453981 |
| 2 | J: 190,000 - 249,999 | 10486.027944111776 |
| 3 | L: 300,000 and above | 10361.04513064133 |
| 4 | G: 130,000 - 149,999 | 6750.503202195791 |
| 5 | H: 150,000 - 169,999 | 6491.765782250686 |
| 6 | I: 170,000 - 189,999 | 6266.973280770915 |
| 7 | D: 70,000 - 89,999 | 5981.598619896492 |
| 8 | F: 110,000 - 129,999 | 5902.4390243902435 |
| 9 | E: 90,000 - 109,999 | 5891.003024193548 |
| 10 | C: 50,000 - 69,999 | 4822.561692126909 |
| 11 | B: 30,000 - 49,999 | 2312.5904486251807 |
| 12 | A: Below 30,000 | 1537.0722433460076 |
| 13 | *(null)* | 1500 |

```
-- Top 10 customers with highest credit limit
SELECT CUST_ID, CUST_FIRST_NAME, CUST_LAST_NAME, CUST_CREDIT_LIMIT
FROM SH.CUSTOMERS
ORDER BY CUST_CREDIT_LIMIT DESC
FETCH FIRST 10 ROWS ONLY;
```

OUTPUT

| | CUST_ID | CUST_FIRST_NAME | CUST_LAST_NAME | CUST_CREDIT_LIMI |
|---|---|---|---|---|
| 1 | 36117 | Abner | Everett | 15000 |
| 2 | 28117 | Abner | Kenney | 15000 |
| 3 | 17450 | Abner | Kenney | 15000 |
| 4 | 6784 | Abner | Kenney | 15000 |
| 5 | 46117 | Abner | Kenney | 15000 |
| 6 | 38784 | Abner | Kenney | 15000 |
| 7 | 25451 | Abner | Everett | 15000 |
| 8 | 14784 | Abner | Everett | 15000 |
| 9 | 4117 | Abner | Everett | 15000 |
| 10 | 25470 | Abner | Everett | 15000 |

```
------------------------------------------------------------
-- Relationships with Tables
------------------------------------------------------------
-- Sales by customer income level
SELECT c.CUST_INCOME_LEVEL, SUM(s.AMOUNT_SOLD) AS total_sales
```

```sql
FROM SH.SALES s
JOIN SH.CUSTOMERS c ON s.CUST_ID = c.CUST_ID
GROUP BY c.CUST_INCOME_LEVEL
ORDER BY total_sales DESC;
```

OUTPUT

| | CUST_INCOME_LEVEL | TOTAL_SALES |
|---|---|---|
| 1 | F: 110,000 - 129,999 | 18433258.55 |
| 2 | E: 90,000 - 109,999 | 14060378.18 |
| 3 | D: 70,000 - 89,999 | 10808681.69 |
| 4 | G: 130,000 - 149,999 | 10152940.96 |
| 5 | H: 150,000 - 169,999 | 9139531.38 |
| 6 | C: 50,000 - 69,999 | 8530084.43 |
| 7 | I: 170,000 - 189,999 | 6927546.12 |
| 8 | B: 30,000 - 49,999 | 5627453.66 |
| 9 | A: Below 30,000 | 4185856.19 |
| 10 | J: 190,000 - 249,999 | 4107370.63 |
| 11 | K: 250,000 - 299,999 | 3588728.07 |

```sql
-- Customers by Country
SELECT co.COUNTRY_NAME, COUNT(*) AS customer_count
FROM SH.CUSTOMERS cu
JOIN SH.COUNTRIES co ON cu.COUNTRY_ID = co.COUNTRY_ID
GROUP BY co.COUNTRY_NAME
ORDER BY customer_count DESC;
```

OUTPUT

| | COUNTRY_NAME | CUSTOMER_COUNT |
|---|---|---|
| 1 | United States of Ame | 18520 |
| 2 | Germany | 8173 |
| 3 | Italy | 7780 |
| 4 | United Kingdom | 7557 |
| 5 | France | 3833 |
| 6 | Spain | 2039 |
| 7 | Canada | 2010 |
| 8 | Brazil | 832 |
| 9 | Australia | 831 |
| 10 | China | 712 |
| 11 | Poland | 708 |

| | COUNTRY_NAME | CUSTOMER_COUNT |
|---|---|---|
| 9 | Australia | 831 |
| 10 | China | 712 |
| 11 | Poland | 708 |
| 12 | Japan | 624 |
| 13 | Singapore | 597 |
| 14 | Argentina | 403 |
| 15 | Denmark | 383 |
| 16 | New Zealand | 244 |
| 17 | Turkey | 91 |
| 18 | South Africa | 88 |
| 19 | Saudi Arabia | 75 |

```
-- Spending by Age Group
SELECT
    FLOOR((EXTRACT(YEAR FROM SYSDATE) - cu.CUST_YEAR_OF_BIRTH)/10)*10 AS age_group,
    SUM(s.AMOUNT_SOLD) AS total_spent
FROM SH.SALES s
JOIN SH.CUSTOMERS cu ON s.CUST_ID = cu.CUST_ID
GROUP BY FLOOR((EXTRACT(YEAR FROM SYSDATE) - cu.CUST_YEAR_OF_BIRTH)/10)*10
ORDER BY age_group;
```

OUTPUT

| | AGE_GROUP | TOTAL_SPENT |
|---|---|---|
| 1 | 20 | 1275035.78 |
| 2 | 30 | 12426239.61 |
| 3 | 40 | 14824746.88 |
| 4 | 50 | 18493693.07 |
| 5 | 60 | 25145489.68 |
| 6 | 70 | 16752652.63 |
| 7 | 80 | 7398042.09 |
| 8 | 90 | 1889931.47 |